KRUSHAK ODISHA

Odisha's Farmer Database

DETAILED PROJECT REPORT (DPR)-PHASE-II

(INTERVENTION UNDER NeGP-A ACTION PLAN-2021-22)

Department of Agriculture and Farmers' Empowerment

GOVERNMENT OF ODISHA

1. Background

Odisha is a state in which majority of its population is dependent on agriculture for livelihood. It is essential that the state has a transparent view of all farmers of the state including cultivators, labourers and sharecroppers. This is important for various reasons, from streamlining schemes and service delivery to long-term farmers' welfare, among others.

Therefore, while the need for a transparent and holistic view exists and cannot be questioned, existing systems and databases developed as part of various projects have only been able to create siloed lists of farmers from all over the state.

To consolidate these lists, preliminary steps were taken as part of the KALIA (Krushak Assistance for Livelihood and Income Augmentation) Scheme launched by the Government of Odisha in December, 2018. As a result of the extensive data unification and verification exercises undertaken for preparation of the final beneficiary lists under KALIA, the state has been able to create an authentic database of roughly 50 lakh Small and Marginal Farmers and Landless Agricultural Households.

To further leverage this base farmer list prepared under KALIA, the Department of Agriculture and Farmers' Empowerment (DA&FE), Government of Odisha has been entrusted with the task of creation of a Unified Farmers' Database for the state as one of the activities outlined under the Hon'ble Chief Minister's vision of 5T – Technology, Teamwork, Transparency, Time and Transformation.

The data set, so developed will act as a bridge and will supplement the effort of GOI for development of a federated data base of the land holding farmers and consequent feeder to the IDEA tech-architecture.

2. Document Scope & Objective

This document is a detailed project report for building Odisha's Farmer Database (OFD).

Topics Covered

The document is divided into the following four parts –

- **Problem Statement:** This part explains the problem statement that is being solved
- **Vision and Use cases**: This part explains the broad vision and use cases that will be enabled till April, 2025
- **Scope and Concepts**: This part explains the concept framework along with relevant tech stack for development of the farmer database
- **Implementation Plan**: This part explains the phased-out development and rollout plan till April 2025
- **Financial Requirement**: This part explains the estimated financial requirement for the rolling out of the plan .

3. Problem Statement

Considering existing Farmer ID and other siloed dataset in Odisha, in the current scenario there are multiple challenges faced by both the primary stakeholders i.e. farmers and the state.

Some of the broader challenges that farmers' face are as follows:

- **Multiple Farmer Registration Platforms (Farmer ID/PMFBY/P-PAS/MIDH/PMKSY/RAD) –** There are multiple platforms on which a farmer registers today. Each portal generates a separate identification number for the farmer. Not only does this lead to an additional hassle for the farmer but also has led to the creation of information silos and duplicate datasets.

- **Lack of recognition of landless and tenant farmers –** The current databases do not recognize a large chunk of sharecroppers and landless labourers, and even if they do, such categorization is not recorded in the system. Through KALIA, the state made its first attempt in this direction.

- **No centralized mechanism to check eligibility for a particular scheme/service –** There are 95 schemes for farmers, currently operational in Odisha including Central Sector, Centrally Sponsored and State Plan schemes. It is extremely difficult for farmers to check their eligibility for any of these schemes given the absence of a proper mechanism.

- **Low chances of customized advisory due to incomplete information –** The absence of a single database with the contact details of all farmers prevents the services such as Ama Krushi and m-Kisan being implemented comprehensively. This prevents farmers from receiving customized, on-time and valuable advisory.

In a similar way, there are challenges faced by the state:

- **Duplicate/Ghost records due to poorly verified and disconnected databases –** In the past few years, the state has taken multiple steps to digitize information sets, however that has led to creation of information silos and therefore, duplication of records. There is a need to verify and consolidate all the available datasets into a unified database so that it can be utilized by all the departments.

- **No mechanism to estimate the number of beneficiaries for a scheme –** Prior to KALIA, there was no data-backed mechanism for the state to estimate the number of beneficiaries eligible under a particular scheme. Each registration was driven by proofs submitted by the farmer and thereafter verified by the relevant government official. This entire process was time-consuming, costly and still left a substantial room for error.

- **Low usage of farmer level insights in policy formulation and decision making –** Because of unavailability of consolidated farmer level information at the state level, there is a potential gap in state formulated agricultural policies.

Please note, the above-mentioned list of challenges is not exhaustive and is only indicative.

4. Vision

The vision of Odisha's Farmer Database is to create a golden record of farmers in the state to solve the problems outlined above and enable the following use cases –

4.1. Identity Proof

Enrolment of all the farmers in Odisha onto a unified platform, will reduce a lot of hassles for farmers while availing different schemes and services. A **Farmer ID card with a secure QR code** will be generated for all farmers to streamline processes, and enable a hassle-free experience for them. It will also prevent any leakages in the system.

4.2. Digital Governance (Integration with land records and other farmer systems)

To avail any government scheme/service, there are multiple documents that are required to be submitted/uploaded by the citizen/farmer as proofs. It is important to note that most of these documents that are required as proofs, are often owned by other government departments only. However, the two systems exist in silos and in most cases, do not interact.

On the other hand, **the triangulation of data (interconnected data systems) is a very powerful tool, to check for anomalies/fraud detection**.

On similar lines, in the current scenario, the verification/authentication of beneficiaries during service delivery is largely left to the ability of field officers. This creates a lot of room for error. **Farmer authentication through Image recognition is being planned for a robust authentication mechanism and to prevent any leakages in the system.**

Therefore, on one hand to make the process hassle-free for end beneficiaries and on the other to enable effective fraud/anomaly detection, it is important that an interconnected ecosystem be created, in which all relevant data systems co-create, co-exist and at all points in time interact.

Odisha's farmer database will be connected to all the existing systems at state and central levels, especially Bhulekh - Odisha's land records and the National Farmer Database. Thus, an interconnected ecosystem will be created to enable triangulation of data and therefore, digital governance.

4.3. Scheme Dissemination

Across the three directorates of the Department of Agriculture, Government of Odisha, there are 80+ schemes across the year. It is important to track the delivery of these schemes and record progress/achievement in real-time. This would not only help in getting a transparent view of the current status but also help in taking corrective action, wherever required.

Odisha's farmer database, supported by DAFE's Decision Support System will enable online target allocation and real-time achievement submission across levels, making review and monitoring of progress transparent and data-backed.

4.4. Geo-tagging, Geo-fencing and Beneficiary tagging

The entire agricultural extension system is involved in last mile delivery of various schemes, including state and central. The Government of Odisha has a mobile application 'Agri-extension App' which is already being used by all the 4000+ extension officers to geo-tag and mark attendance for weekly Gram Panchayat sitting and farmer query resolution. The same application will be linked to Farmer Database and further be enabled with geo-tagging, geo-fencing and beneficiary tagging across schemes for various activities including field demonstrations, plantations, input/minikit distribution, execution of structures, etc.

4.5. Customized Agro-advisory dissemination

While Agriculture is a common phenomenon in India, there are different problems that farmers face on a daily basis. A blanket solution is not going to work for them.

Odisha's Farmer Database will form the base to enable such customized advisory that will be further disseminated by the state's **Ama Krushi** advisory and helpline service.

Scope and Concepts

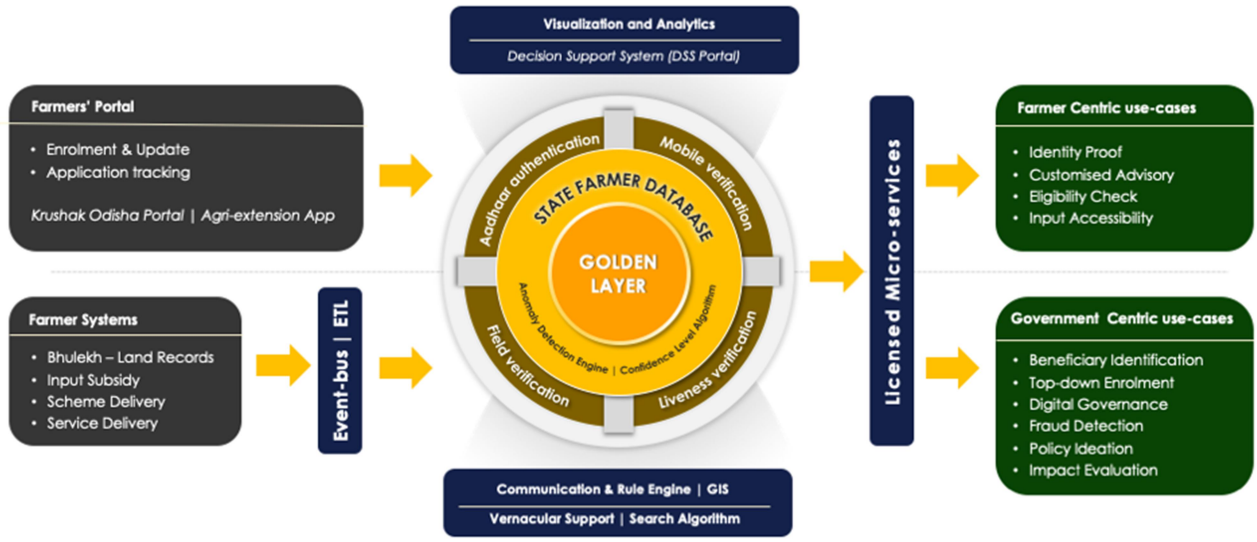
4.6. Design Principles

The following design principles have and will guide the creation of the OFD's technical architecture.

- Creation of modular, reusable, scalable, extensible & deployable technology products using open-source tools
    a. **Modular** - Can be configured by wiring together multiple open source tools
    b. **Reusable** - Can be reused with minimal effort for similar use cases
    c. **Scalable** - Can work well as number of end users begin to rapidly scale
    d. **Extensible** - Feature additions can be made on the products with ease
- To leverage and integrate with existing government systems

4.7. OFD Framework

Keeping the design principles as a base, the overall framework diagram for OFD is as shown below -



As per the framework, the entire concept of OFD can be divided into various components -

- **State Farmer Database**: The central circle of State Farmer Database, with an external four part verification layer and internal Confidence Level algorithm and Anomaly Detection engine
- **Farmer's Portal**: Krushak Odisha web portal and Agri-extension mobile application for enrolment and verification
- **Farmer Systems**: Integration with Bhulekh (Land Records) and all the other existing transactional databases
- **Features and Enhancements:** Communication & rule engine, GIS, Vernacular support, search algorithm, visualisation and analytics will be enabled through Department's Decision Support System (DSS) Platform
- **Micro-services**: Licensed API-services for enabling use-cases

4.8. Definition

To build a robust Farmer Database, it is essential to define an individual farmer.

For the purposes of Odisha's Farmers' Database, a farmer would include cultivators (including sharecroppers) and agricultural laborers, who may or may not own land, and are engaged in any of the following activities - Crop Production, Animal Production, Forestry and Fishing/Aquaculture.

4.9. Database Record Types

OFD will store two types of records –

- **Individual data -** OFD will store individual farmers' data. Aadhaar will be used to uniquely identify an individual.
- **Family structure -** OFD will group farmers into a family and identify their relationships with each other. An internally generated 'family ID*' (or Ration no. if found in NFSA Database) will identify whether all farmers are part of one family while 'position in family tree' attribute will determine his/her relationships with others.

4.10. Database Attributes

After the analysis of farmer registration forms for schemes and services such as Farmer ID, KALIA, PM KISAN, m-Kisan and PMFBY, it was observed that **most of these forms have similar fields**, except few fields relevant to the specific scheme.

OFD will store the following attributes of a farmer ***(Details in Annexure I)*** –

#	Attribute	#	Attributes
1	**Farmer Name**	12	**Residential Address**
2	**Gender**	13	**Assets - Land Details including geo-coordinates**
3	**Date of Birth / Year of Birth**	14	**Assets - Crop Details**
4	**Community**	15	**Whether Sharecropper**
5	**Marital Status**	16	**Other Profession**
6	**Mobile number**	17	**Bank Details**
7	**Email ID**	18	**Aadhaar No.**
8	**Photograph**	19	**Farmer ID**
9	**Relative Name**	20	**Family ID**
10	**Relation with Farmer**	21	**Position in family tree**
11	**Relative Aadhaar No.**	22	**Farmer Consent**

Please note, new assets (similar to land and crops) can be added in future

4.11. OFD ID Format

Please note, for all the landowning farmers i.e. part of the National Farmer Database, a centralised farmer ID as generated by the Ministry of Agriculture and Farmers' Welfare, Government of India will be used.

Only in scenarios, such as for landless farmers, who are not covered under the National Farmer Database a state ID will be generated, in the following format - a 10 digit randomised alpha-numeric code.

4.12. OFD ID Revision Principles

Please note, once a farmer is allotted a unique ID, that ID cannot be revised/altered at any point in time in the future. In cases where a particular farmer expires/becomes ineligible, the same information may be recorded against that particular ID.

4.13. OFD ID Card Components

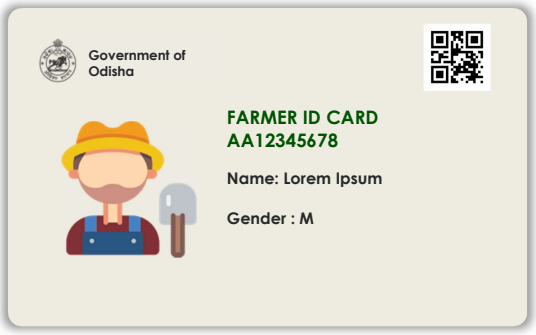
Following principles will govern the generation of digital copies of OFD ID Cards

- To enable offline verification in the future, all ID Cards to have a secure QR code that can be scanned by the department QR code scanning application only
- The QR code would store basic and static information of the farmer such as OFD ID, Farmer name*, Gender, Date of Birth and Photograph

- Only static information is stored to prevent further generation of QR code
- In case an existing static data-point is recorded incorrectly and therefore is updated by the farmer, a new QR code will be generated
    - Farmer will have to take a new print out of his/her Farmer ID card
    - The exercise will also include discarding the older QR code so that it becomes non-functional for any further verifications

**Farmer Name, even though a slow-moving field has been added to the QR Code because it is fundamental to farmer identification*

A sample design of the Farmer ID card is shown below –



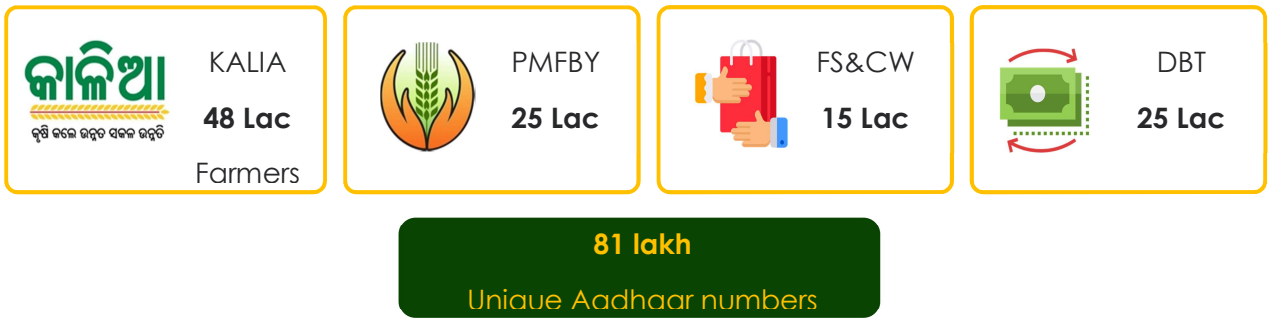
4.14. Database Connection and Sync Strategy

In order to prevent the creation of information silos in the future, and cover all the multi-dimensional aspects of a farmer profile, it is important to **create an integrated tech system** i.e. link existing and relevant datasets to the OFD.

There are four types of databases relevant in this context –

- **Registered list of farmers** - Presently several farmer lists exist under various systems such as Farmer ID, KALIA, PMFBY, P-PAS, M-PAS, PMKSY, etc. Four farmer lists* have been shortlisted namely *KALIA, PMFBY, FS&CW and Farmer ID* because of the following reasons –
    - High degree of Aadhaar seeding
    - Most validated farmer lists
    - Cover majority farmers of the state

As highlighted below, these four lists have been consolidated and **a base list of OFD with approx. 81 lakh unique Aadhaar numbers** has been prepared.



*Please note, in future, other farmer lists can be uploaded in a bulk manner

- **Bhulekh (Land Records) -** Odisha's land records i.e. Bhulekh will be linked to the Farmer Database to authenticate and maintain the actual ownership and size of landholdings. Preliminary work on this has already been done for PMFBY (Pradhan Mantri Fasal Bima Yojana) Registration, where the land records have been integrated with PMFBY portal. This will further be expanded, to ensure all landowning farmers have land details recorded that are linked to and verified from Bhulekh.
- **Transactional data -** Each recorded farmer has been transacting with entities in the agriculture ecosystem for activities such as purchase of inputs, sale of final produce, etc. However, currently all these transactions are recorded in different databases.
    - ***For Farmer authentication*** - All these transactional systems will be linked to the main farmer database for farmer authentication while transactions are being executed.
    - This authentication will happen through Aadhaar eKYC in the online environment and Face Recognition/Match (stored in the QR Code of the Farmer ID card) in the offline environment
    - ***Smart-Contracts -*** Odisha government has recently rolled out the BALARAM scheme (Bhoomihina Agriculturist Loan and Resources Augmentation Model) to finance tenant farmers on the basis of cultivation certificates. These cultivation certificates will be linked to OFD and will be converted into smart contracts in the future to enable faster service dissemination.
        - On similar lines, in the future other such G2F (Government to Farmer) agreements are planned to be done through smart contracts

- **Authentication Ecosystem -** All farmer records will go through a layer of authentication before they are added to the farmer database. While some of the following checks are mandatory for inclusion in the database, others are optional -
    - Aadhaar Demographic/Biometric Authentication (Mandatory)
    - Field verification (Mandatory)
    - Mobile verification (Optional)
    - Liveness verification (Optional during enrolment; Mandatory for transactions)

- **Occupational data -** A farmer may be a main worker or marginal worker, the implication of this being that he/she may take up additional occupations/professions apart from farming during any financial year.
    - This data point is relevant for creation of beneficiary lists under various schemes, where such other occupations are considered as exclusions
    - For this purpose, OFD will record other occupation of all the farmers and this will be maintained through a combination of existing databases and field verification
    - The following state databases will be linked to capture the other occupations of all farmers -

        - HRMS
        - Teachers
        - Police

- **National Farmer Database - The state farmer database will be integrated with the National Federated Farmer Database through micro-services to be provided by the state and consumed by the central team. This is to ensure that there is no duplication of work at state and central levels, and there is seamless flow of data between the two entities. Additionally, Farmer ID generated for the landowning farmers can be synced with the state database. Micro-services are explained in C2.10.**

4.15. Micro-services for Use-Case Enablement

To enable OFD use-cases, the following micro-services will be made available:

- Access to farmer data or/and aggregate counts as defined in the licensing agreement
- Access specific to geography defined in the licensing agreement
- A set of pre-defined/suggested data-points combination available to be selected from
    - Rule engine to further customise/create new access as per requirement

4.16. Micro-services Licensing

- Each license will be tied down to:
    - One entity
    - A set of applications
    - A list of use-cases for which data will be used
- A user's license will determine:
    - The micro-services are available to the user
    - The data within a micro-service available to the user
    - Usage-limit for each micro-service, if applicable
- The scope of a license can be revised, whenever required

4.17. Micro-services Terms-of-Usage

- Micro-services can only be used
    - by licenses bodies
    - for licensed application
    - for licensed use-cases
- Micro-services cannot be used for policing purposes

4.18. Technology Components

The following tech applications/components will be used for the creation of Odisha's Farmer Database -

#	Component	Application	Description
1	**PostgreSQL (PSQL)**	Database Management	The entire farmer database is maintained on this open-source PSQL application
2	**NodeJS**	Database Models and Migrations	• To create models for defining data validations • For any future changes in the database schema
3	**ReactAdmin**	Farmers' Portal	Open-source React Admin component is being used for creation of a Farmers' Portal
4	**FusionAuth**	Access Control (ACL)	An open-source and scalable access control layer to authenticate and authorize different users on the basis of available/tagged licenses (data access)
5	**Graph QL and Hasura**	Configuring APIs	Open-source components to enable configurable APIs through the front-end interface. This is being used to control access to different types of users.
6	**Talend + Airflow**	Configurable rule engine	Open-source ETL(Extract-Transform-Load) tool for development of a configurable rule engine for defining confidence level algorithm, anomaly detection engine and schemes eligibility engine
7	**Drools**	Configurable rule engine	Open-source rule engine to define simpler business logics and rules
8	**Secure QR Code**	Ease in offline service delivery	A secure QR code is being generated to enable smooth and hassle free registration for government services in the offline environment
9	**Face Recognition / Face Match**	Offline farmer authentication and liveness check	A Face match API will be integrated to enable offline authentication of farmers during various transactions with the government
10	**Phonetic Search**	Enhanced search ability in the database	Not all readily recall their identity numbers including Aadhaar, etc. Additionally, the same name is spelt in different ways by different people warranting requirements for an advanced search mechanism.
11	**AI enabled WhatsApp Chatbot**	AI enabled customized Agro-advisory	An AI enabled whatsapp chatbot for farmer query resolution and customized agro-advisory

Along with these components, a combination of

- Krushak Odisha Web-portal (Unified Farmers' Portal) and
- Mobile app (existing Agri-extension mobile app being used by 4000+ Gram Panchayat level officers)
- Decision Support System (DSS) Portal

will be used for development, management and verification of the farmer database.

5. Implementation Plan

#	Description	Aug '21	Oct '21	Apr '22	Apr'22-23	Apr'23-24	Apr'24-25
a)	Study of Existing System & Processes and Finalization of SRS						
b)	Design & Development of Krushak Odisha Portal						
c)	Geo-tagging, geo-fencing and beneficiary tagging for various schemes						
d)	Integration with Millet Procurement System (M-PAS) & Integration with Paddy Procurement System (P-PAS)						
e)	Integration with Bhulekh - Land records						
f)	Integration with Decision Support System - Reporting, Visualisation and Analytics						
g)	Go-live of Krushak Odisha Portal after Security Audit						
h)	Operation & Maintenance						
i)	Technical Support Unit						

6. Reference to the DPR PHASE-I submitted to GOI earlier and progress there of:

To meet different ancillary expenditures for the development of the said data base, initially Govt. of India have been moved with a DPR with project cost of 14.7954 crore for 5 years i.e up to 2024-25 with a request for sanctioning the funds from the NeGP-A action plan of 2020-21.But as required by GOI ,the said DPR was recast with a project cost of 2.9733 crore to be expended during 2020-21 with the following provisions.

Cost Estimates for FY 2020-21		
#	**Particulars**	**Cost (in INR)**
1	Development, Customization and System Integration	1,40,00,000
2	Consultancy > Software (Domain Expert-SME)	15,00,000
3	Hardware IT Infrastructure	25,00,000
4	Software Support/Technical Support Unit (TSU)	36,00,000
5	Social Media/PR Support	10,00,000
6	Helpdesk Support	60,00,000

7	Web Security Audit	5,00,000
8	SSL Certificate	50,000
	Sub-total	**2,91,50,000**
9	Administrative Cost (2% of project cost)	5,83,000
	Grand Total	**2,97,33,000**

The GOI share of 60% was received during last week of March-21. Accordingly, the System Support Team have already been onboarded through tender process conducted by E&IT Department of Odisha with a total cost of 4,91,59,809 crore as detailed below with a prerequisite that funds for the purpose should be placed at the E& IT Department at the beginning for onward reimbursement to the selected System Support Team. Now that the entire farmer profiles are being verified by the field functionaries for authenticity.

Sl#	Particulars	Cost			
		Year 01	Year 02	Year 03	Total
a)	Development, Customization and System Integration	2,05,05,050	-	-	2,05,05,050
b)	Operation & Maintenance	30,05,050	25,50,550	25,50,550	81,06,150
c)	Technical Support Unit (TSU)	29,64,000	29,64,000	29,64,000	88,92,000
d)	SSL Certificate (3 Years Validity)	70,005	-	-	70,005
e)	Security Audit	1,00,100	1,00,100	1,00,100	3,00,300
	Total (Excluding Consultancy Fees & Taxes)	**2,66,44,205**	**56,14,650**	**56,14,650**	**3,78,73,505**
	Consultancy Fees @10%	**26,64,421**	**5,61,465**	**5,61,465**	**37,87,351**
	Sub-Total (Excluding Taxes)	**2,93,08,626**	**61,76,115**	**61,76,115**	**4,16,60,856**
	GST @18%	**52,75,553**	**11,11,701**	**11,11,701**	**74,98,954**
	Grand Total	**3,45,84,178**	**72,87,816**	**72,87,816**	**4,91,59,809**

7. Financial Requirement for DPR PHASE-II

Considering the receipt of Rs 2.9733 crore from GOI during 2020-21, expenditure under different heads and to meet the further cost towards reimbursement for the SST as per the above-mentioned table to be placed at E&IT Department during 2021-22, the exact revised financial requirement can be summed up as follows.

Cost Estimates for DPR-PHASE-II		
#	Particulars	Cost (in INR)
1	Funds towards Development, Customization and System Integration	2,75,09800
2	Change request in future (Tentative)	50,00,000
	Sub-total	**3,25,09800**
9	Administrative Cost (2% of project cost)	6,50,100
	Grand Total	**3,31,59,900**

8. Outcomes of the project

(A)

1. Identification of Beneficiaries with identity proof
2. Instant Eligibility Check
3. Input Accessibility
4. Customized Agro-advisory

Farmer centric

5. Fraud detection
6. Policy Formulation and
7. Impact Assessment
8. Data backed planning
9. Digital Governance
10. Proactive service delivery

Govt. centric

(B)

1. The farmer data base , so developed will interact with the Federated data base developed by GOI.
2. It will also form the State core in the IDEA architecture created by GOI.
3. Integration with different farmer centric applications, information portals like “Ama krushi”